TITLE PAGE

- **Problem Statement ID –** SIH26090
- **Problem Statement Title-** AI-Driven Market Linkage and Smart Cataloging App for Marginalized Artisans
- **Theme-** Heritage & Culture
- **PS Category-** Software
- **Team ID-** P49
- **Team Name-** Srijan



Problem:

Marginalized artisans depend on **seasonal fairs** for income. Lack of **year-round online selling** and **digital skills** limits their **customers, market reach, and earnings**.

Our Idea :

Our **AI-powered mobile app** turns an artisan's **phone camera into an online shop**. With just one photo, AI **improves the image, creates the product listing, suggests the right price**, and helps sell products **throughout the year**.

Proposed Solution :

- **AI Image Studio:** Automatically **removes background clutter, improves lighting, and enhances product photos** for online selling.
- **Multilingual Auto-Cataloger:** Artisans can **describe products through voice in regional languages**, while AI creates **Hindi and English product descriptions**.
- **Dynamic Pricing Assistant:** Suggests a suitable selling price using **raw material costs, product information, and current market trends**.
- **Digital Storefront & Sales Analytics:** Makes products available through a **year-round online marketplace** with a **simple dashboard** showing **orders, earnings, and demand trends**, reducing dependence on **physical exhibitions**.

Innovation/Uniqueness:

- **Voice-First E-Commerce Experience:** Artisans can create product listings by **simply speaking** instead of **typing or manually filling complex e-commerce forms**.
- **Regional Language to Market-Ready Content:** Transforms **regional-language descriptions** into **professional Hindi & English content**, while preserving the artisan's **unique product story**.
- **Artisan-Centric Design:** Hides complex tasks behind a **simple, low-literacy-friendly interface**, letting artisans focus on **their craftsmanship**.

Technology Stack:

Frontend : Flutter/Dart.

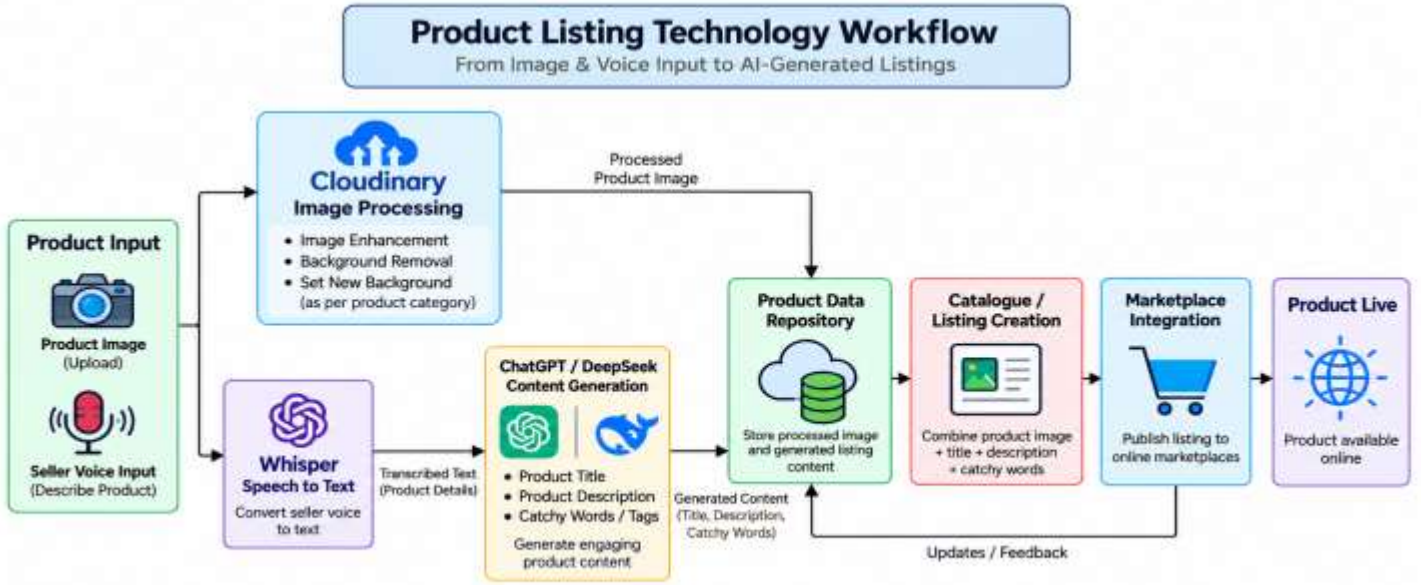
Backend : Node Js, Express Js.

Database : PostgreSQL/Supabase.

AI and ML : Python, OpenAI Whisper, OpenCV, LLM.

API Services : REST API

Deployment : AWS / Render, Docker, GitHub, Git.





FEASIBILITY AND VIABILITY

1. Feasibility

➤ Technical Feasibility

- Uses Cloudinary for image processing.
- Uses Whisper and AI models for content generation.

➤ Operational Feasibility

- Simplifies product listing creation.
- Reduces manual work for artisans.

➤ Financial Feasibility

- Uses affordable cloud-based services.
- Reduces listing and processing costs.

2. Potential Challenges & Security Risks

➤ Data Security & Privacy

- Risk of unauthorized data access.
- Privacy concerns with third-party APIs.

➤ Limited Access to Technology

- Limited digital awareness.

➤ High Traffic

- Server overload during peak seasons.
- Slow AI processing under heavy load.

3. Strategies for Overcoming Challenges

➤ Secure System Architecture

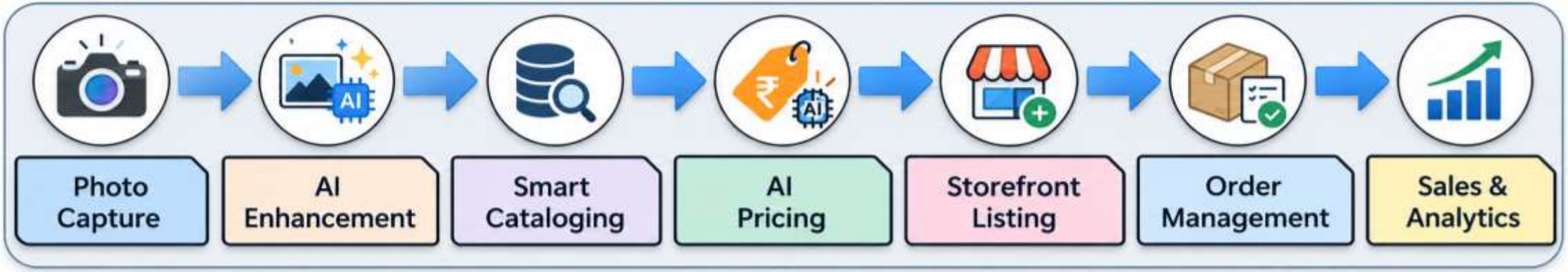
- Use encrypted API connections.
- Apply strong user authentication.

➤ Accessible User Interface

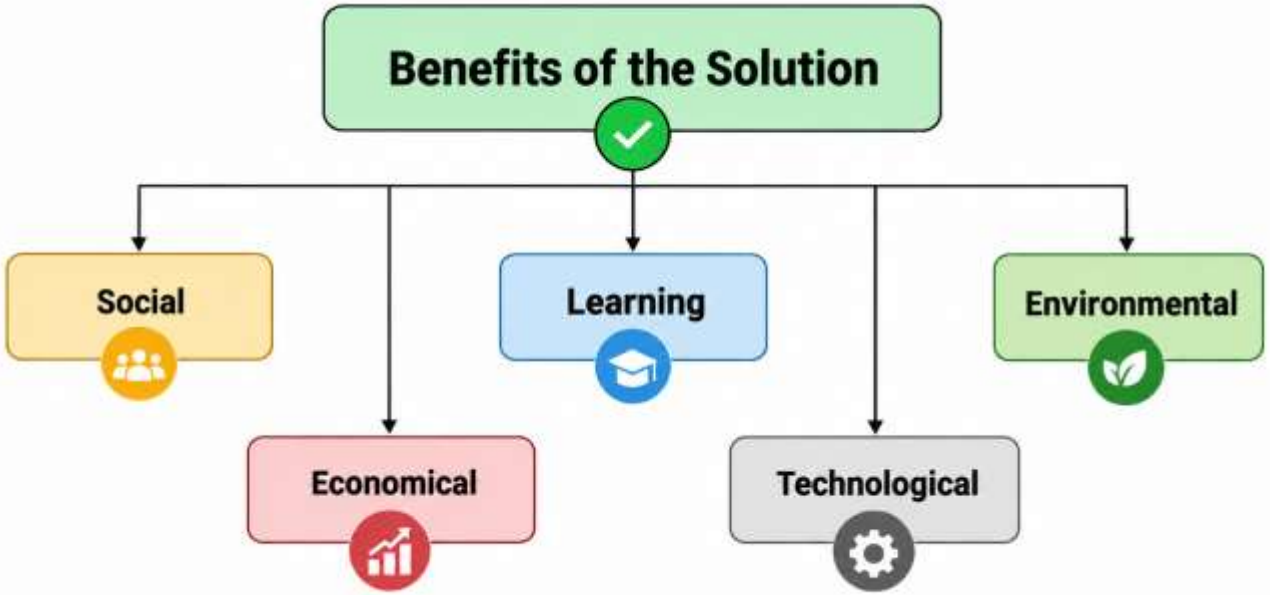
- Provide simple navigation.
- Support regional languages.

➤ Scalable Infrastructure

- Use cloud auto-scaling.
- Optimize image processing.



IMPACT AND BENEFITS



Potential Impact On The Target Audience:

➤ Removes the **digital skill barrier** — artisans no longer need photography, writing, or pricing expertise to sell online.

➤ Cuts out middlemen, so a **larger share of each sale** goes directly to the artisan.

➤ **Voice-first, multilingual design** brings e-commerce within reach of artisans with little to no digital literacy.

➤ Strengthens visibility for **regional and traditional crafts**, connecting them to buyers far beyond their local area.

Benefits Of the Solution:

Social:

- Builds dignity and a connected community for marginalized artisans across regions.
- Preserves traditional crafts by giving them a modern digital platform.

Economical:

- Provides year-round income, reducing dependence on seasonal fairs.
- Increases earnings by cutting out middlemen for fairer pricing.

Environmental:

- Reduces carbon footprint by cutting travel to physical fairs/markets through a digital storefront.
- Promotes low-impact production, as traditional crafts use natural biodegradable materials and low-energy techniques.

Learning:

- Builds digital and financial literacy through guided, repeated use.
- Teaches basic e-commerce skills like cataloging and pricing.

Technological:

- Introduces AI-powered tools to first-time digital users.
- Encourages comfort with mobile technology and future innovation.

RESEARCH AND REFERENCE

References:

- https://socialjustice.gov.in/schemes/13
- https://socialjustice.gov.in/schemes/100
- https://socialjustice.gov.in/schemes/104
- https://socialjustice.gov.in/schemes/99
- https://socialjustice.gov.in/schemes/32

Comparison with Existing Systems

Feature	SRIJAN	TULIP (NBCFDC)	GoCoop / GoSwadeshi	Meesho / Amazon
Phone-Camera Storefront	✓	✗	✗	✗
AI Image Enhancement	✓	✗	✗	✗
Automatic Cataloging	✓	✗	−	✗
AI-Driven Market Pricing	✓	✗	✗	✗
Editable AI Suggestions (Artisan Control)	✓	✗	✗	✗
Voice-Guided, Icon-First Onboarding	✓	✗	✗	✗
Offline-First Capture + Auto-Sync	✓	✗	✗	✗
Multi-Regional Language Support	✓	−	−	✓
E-commerce Integration (Amazon/Flipkart/ONDC)	✓	✓	✗	✓
Data Encryption & Secure Login	✓	✓	✓	✓
Auto-Scaling for Festive Traffic	✓	−	−	✓

✓ Available | ✗ Not Available | − Limited / Basic

Research Workflow

